

END-OF-SEMESTER PROJECT · MLOPS

News Insight

An MLOps platform for Tunisian news analysis

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Agenda

1 Problem & solution

what we built, and why

2 The product

a tour of the working app

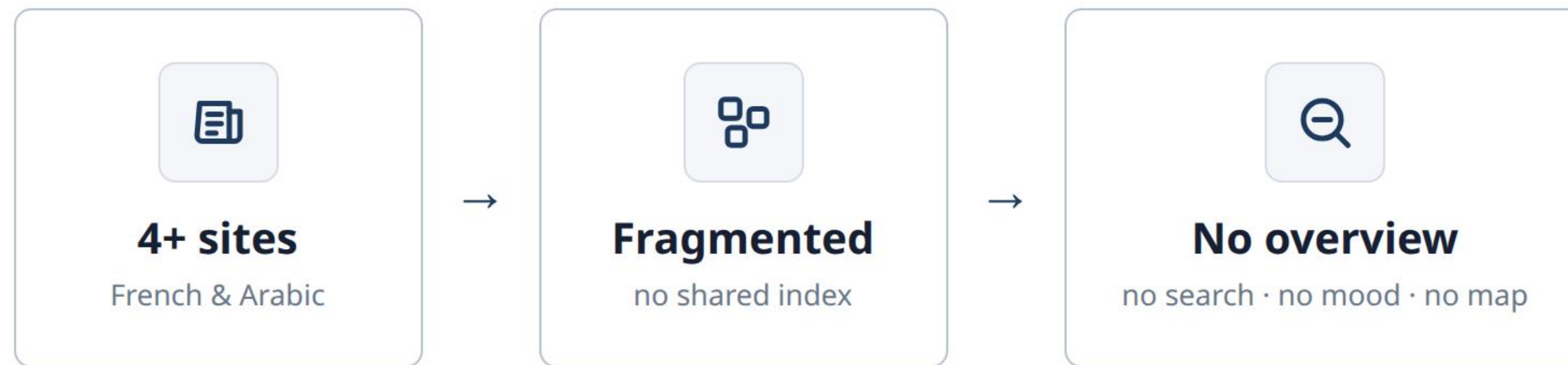
3 Architecture & pipeline

how it works, end to end

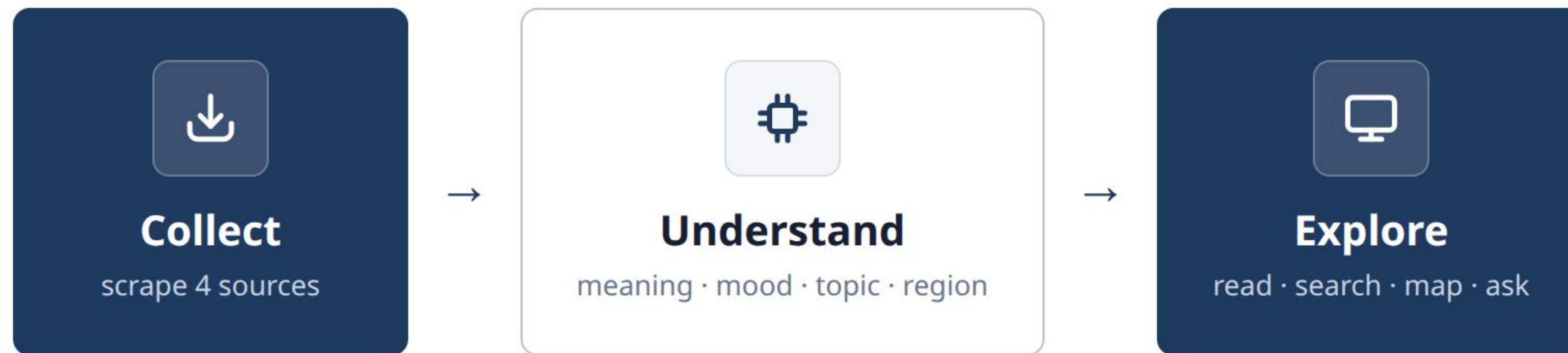
4 Contributions & future work

what we demonstrated, and next steps

Tunisian news is scattered across sites, and hard to search or analyse.



One pipeline collects the news, understands it, and serves it.



Automated end to end and usable from one web app, so no step is manual once it runs.

PART 02

The product

A tour of the working app. The interface is in French, for a Tunisian audience.

Every source in one feed, tagged by topic and mood.

Tuniscop

Lundi 13 Juillet 2026

À la une

Articles

Carte

Recherche

Analyses



SPORT

Wimbledon : La Tchèque Linda Nosková sacrée

mosaïque · il y a 2 j · Positif

Dernières actualités

Toutesmosaïquelapresskapitalis



ENVIRONNEMENT

Vidéo de pollution présumée à la plage “La Sirène” à Hammamet : l’APAL apporte...



ENVIRONNEMENT

Tunisie | La canicule s’installe, comment se protéger ?



SÉCURITÉ

Affaire des enregistrements : Mondher Lounissi reste en prison



ENVIRONNEMENT

Canicule : une seule région tunisienne échappe à l’alerte, le reste du pays sous vigilance

Figure 1. One feed across all four sources; each card shows its source, topic and sentiment.

Sentiment, top topics and trends, at a glance.

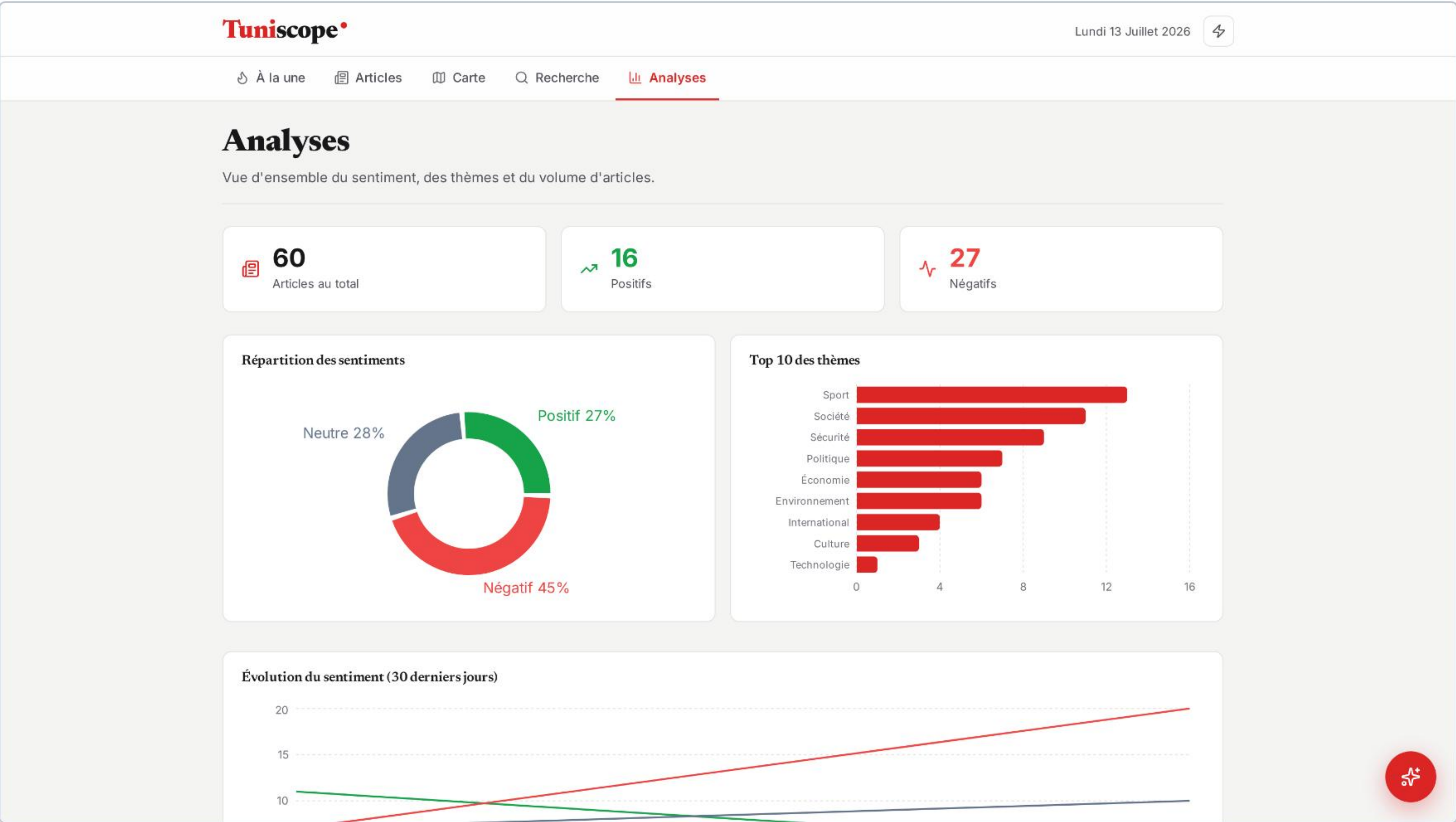


Figure 2. Sentiment split, the most-covered topics, and a 30-day sentiment trend.

Coverage mapped across all 24 governorates.

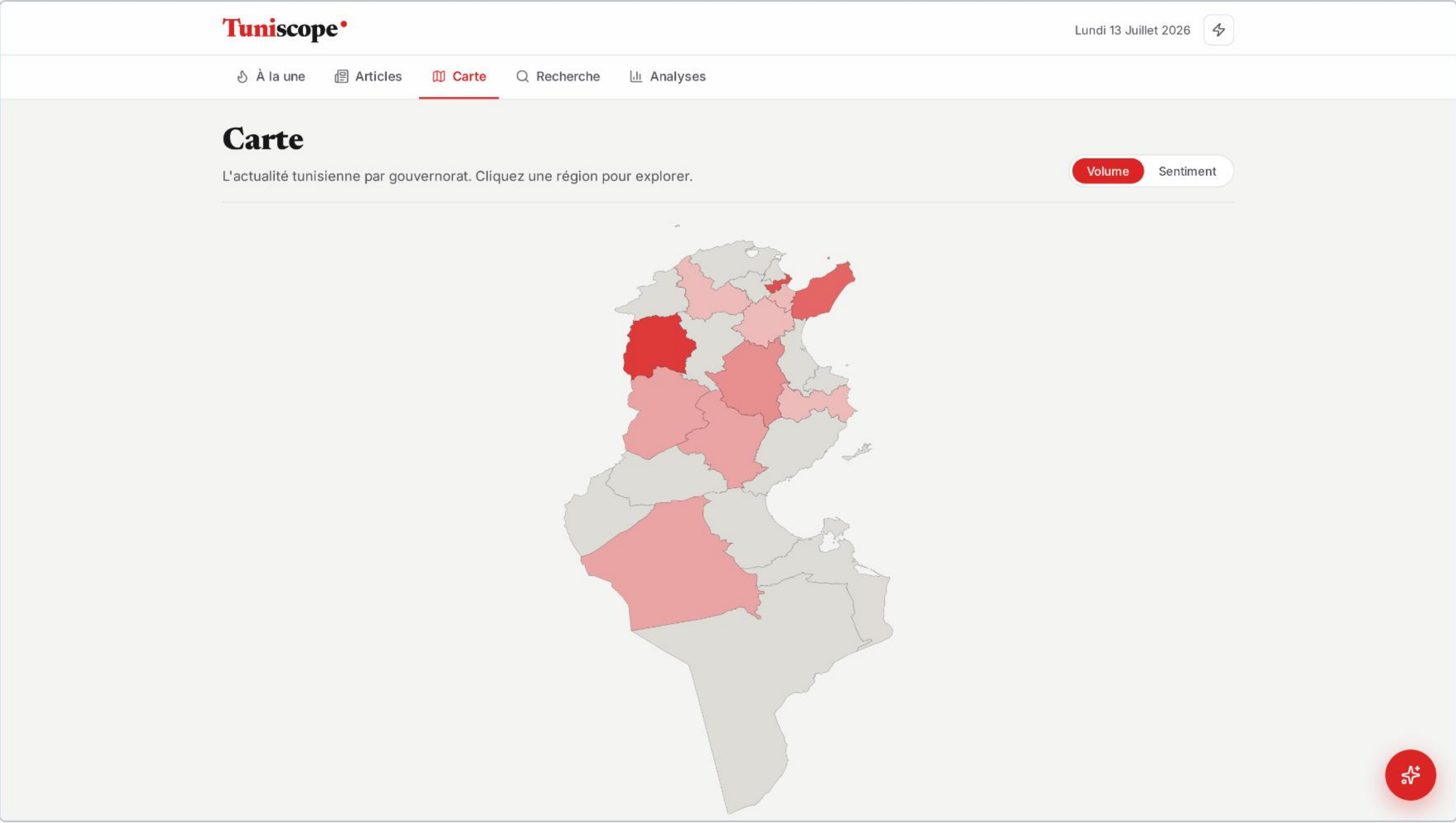


Figure 3. Each governorate is shaded by article volume or by average sentiment.

Search finds articles by meaning, not just keywords.

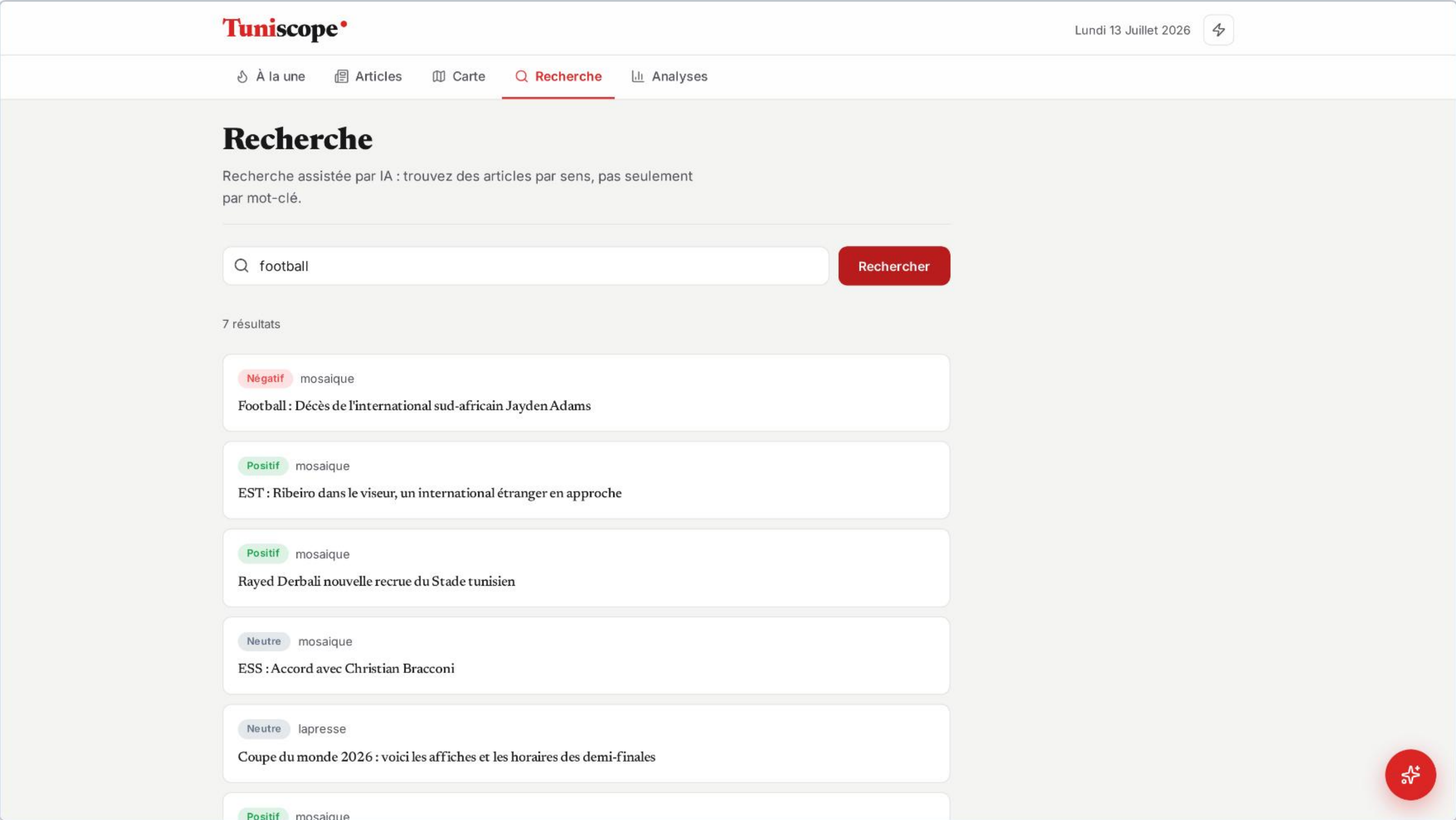


Figure 4. Results are ranked by semantic similarity; each shows its source and sentiment.

The assistant answers from real articles, and cites its sources.

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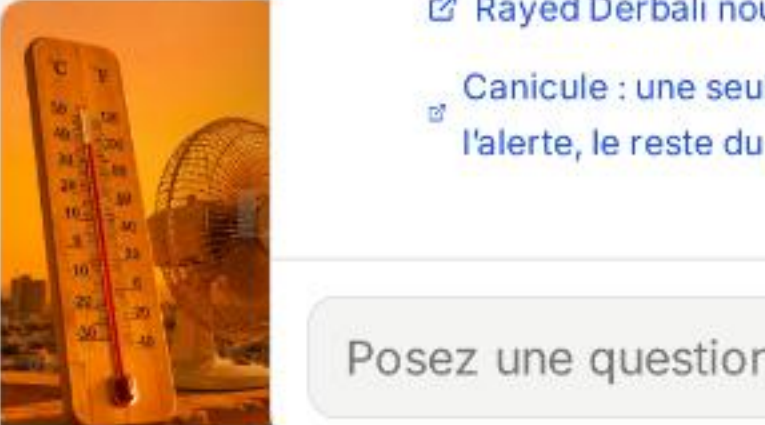
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Assistant IA

Dernières nouvelles sportives ?

D’après les articles fournis, une seule information sportive est disponible : Rayed Derbali est une nouvelle recrue du Stade tunisien, comme indiqué dans l’article [4]. Il n’y a pas d’autres informations sportives dans les articles fournis. [4]

SOURCES

- Tunisie-Russie | Sous le signe de la continuité
- L’Europe finance les intérêts industriels chinois en Tunisie
- Donald Trump, un maître chanteur
- Rayed Derbali nouvelle recrue du Stade tunisien
- Canicule : une seule région tunisienne échappe à l’alerte, le reste du pays sous vigilance

Posez une question...

Figure 5. The answer references the exact articles it used, listed as numbered sources.

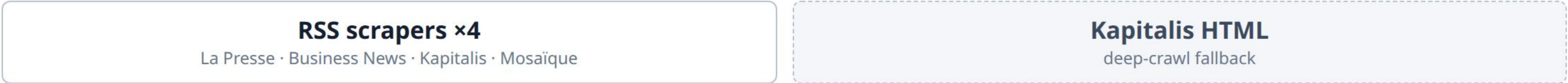
PART 03

Architecture & pipeline

From a web page to a searchable, answerable news feed.

A layered pipeline turns live news sites into a searchable app.

SOURCES



Prefect · scrape → transform → load



ENRICHMENT · reads each row, writes back columns

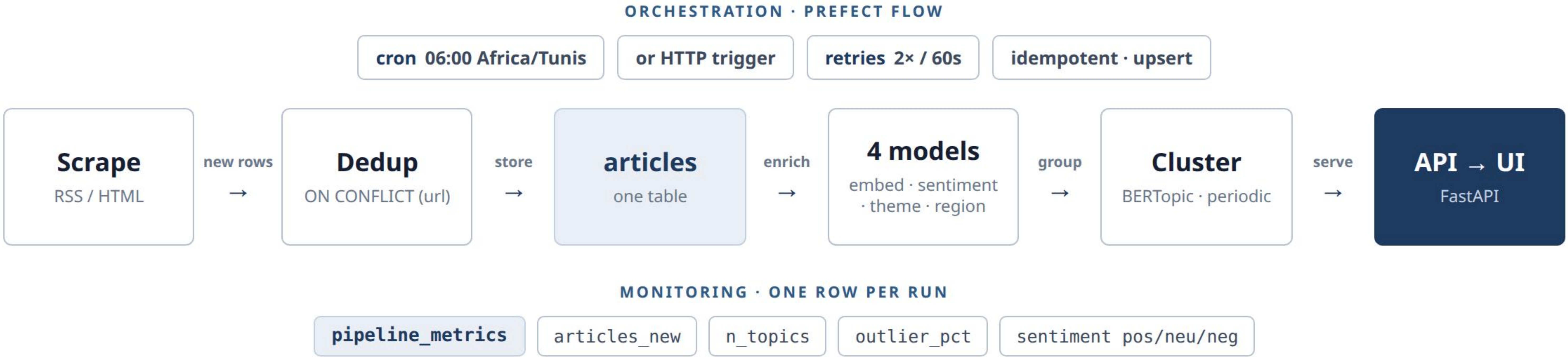


FastAPI serves the enriched data



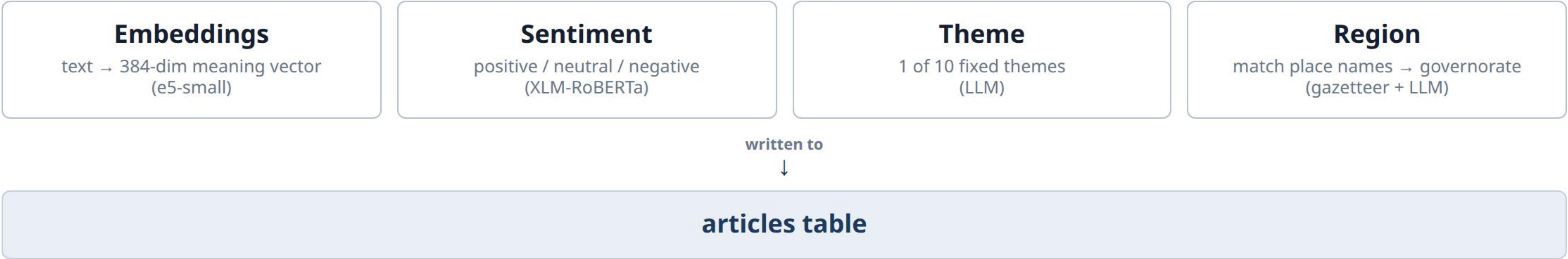
— data flow - - - orchestration / tracking - - - external service

Prefect runs the pipeline on a schedule, and logs every run.



Four models label every article; BERTopic finds the topics.

PER ARTICLE · EACH MODEL WRITES ONE COLUMN



PERIODICALLY · DISCOVER TOPICS ACROSS THE CORPUS



The assistant answers from the 5 nearest articles, not the model's memory.



Retrieval-augmented generation (RAG): the model only sees the retrieved articles, so every answer is grounded in the corpus and can cite exactly where it came from.

Simple to run: one table, one command, all open-source.

DATA MODEL

A single `articles` table: enrichment stored as columns, plus a 384-dim vector for search.

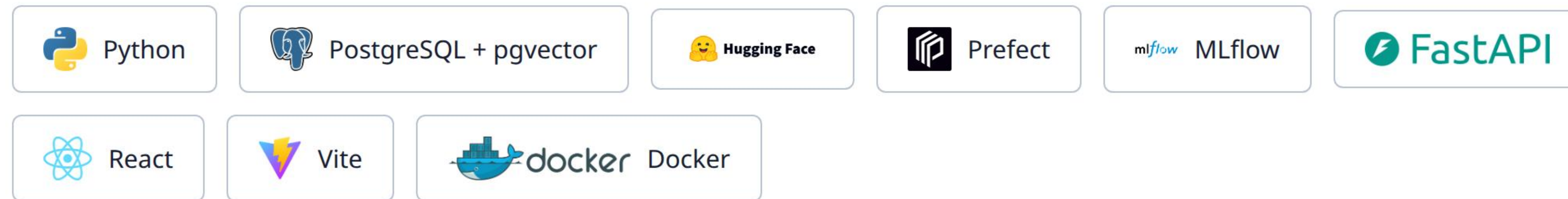
`id` `url` `source` `title` `sentiment` `topic_label` `region` `summary` `embedding vector(384)` `cosine <=>`

DEPLOYMENT

One `docker-compose up` starts five services on one network:

`news_postgres :5432` `news_api :8000` `news_frontend :5173` `news_prefect :4200` `news_mlflow :5000`

STACK · OPEN-SOURCE THROUGHOUT



What this project demonstrates.

1 **A complete end-to-end platform**

From live news sites to a searchable, answerable web app, running today.

2 **Search by meaning**

Semantic retrieval over French and Arabic articles, using pgvector and multilingual embeddings.

3 **An assistant that cites its sources**

Retrieval-augmented answers grounded in the corpus, so nothing is invented.

4 **Reproducible MLOps**

A scheduled, tracked and monitored pipeline; the whole stack starts with one command.

Current limits: four sources and French-first; Arabic dialect and more sources are the next step.

Future work

More sources

beyond the four feeds

Named-entity extraction

people · places · organisations

Arabic dialect

content & sentiment

Continuous scheduler

run the pipeline in production

Thank you

Questions & discussion